

ISA CERTIFIED PERSONAL TRAINER COURSE

WEEK 4 – PRE-PARTICIPATION + FITNESS
ASSESSMENTS + PROGRAM DESIGN 1
(CARDIORESPIRATORY TRAINING)

- CHAPTER 5
PREPARTICIPATION HEALTH SCREENING
- CHAPTER 7
RESTING ASSESSMENTS AND
ANTHROPOMETRIC MEASUREMENTS
- CHAPTER 8
CARDIORESPIRATORY TRAINING: PHYSIOLOGY,
ASSESSMENTS AND PROGRAMMING



CONTENT COVERED



Preparticipation Health Screening



Resting Assessments



Physiology of Cardiovascular and Pulmonary System



Acute Response to Cardiorespiratory Training



Chronic Adaptation to Cardiorespiratory Training



Cardiorespiratory Assessments



CHAPTER 5

PREPARTICIPATION HEALTH SCREENING

Learning Objectives:

- Administer exercise preparticipation health screening for clients
- Recognize scenarios in which a client should receive further evaluation from a qualified healthcare professional prior to engaging in increased physical activity
- Identify additional forms that must be completed by a client to ensure a successful exercise experience
- Explain the effects of select medications on resting and exercise heart rates

PART I. PREPARTICIPATION HEALTH SCREENING

PREPARTICIPATION HEALTH SCREENING

- **Risk factor profiling – may become a barrier to exercise**
- New health screening process: ACSM preparticipation health screening algorithm
- Self-guided screening: PAR-Q+, minimum tool
- ePARmed-X+: for those who has a positive response to PAR-Q+ or requested by a health care professional

PREPARTICIPATION HEALTH SCREENING

LEVEL OF
PHYSICAL
ACTIVITY

DIAGNOSED CV,
METABOLIC OR
RENAL DISEASE

DESIRED
EXERCISE
INTENSITY



- Pain or discomfort in the chest (angina), neck, jaw, arms, or other areas that may result from myocardial ischemia
- Dyspnea: shortness of breath at rest or with mild exertion
- Orthopnea: dyspnea in a reclined position
- Paroxysmal nocturnal dyspnea: onset of dyspnea 2-5 h after sleep
- Dizziness or syncope: commonly caused by reduced perfusion to the brain
- Ankle edema: swelling due to excess fluid
- Palpitations or tachycardia: forceful or rapid heartbeat



ACSM PREPARTICIPATION HEALTH SCREEN ALGORITHM

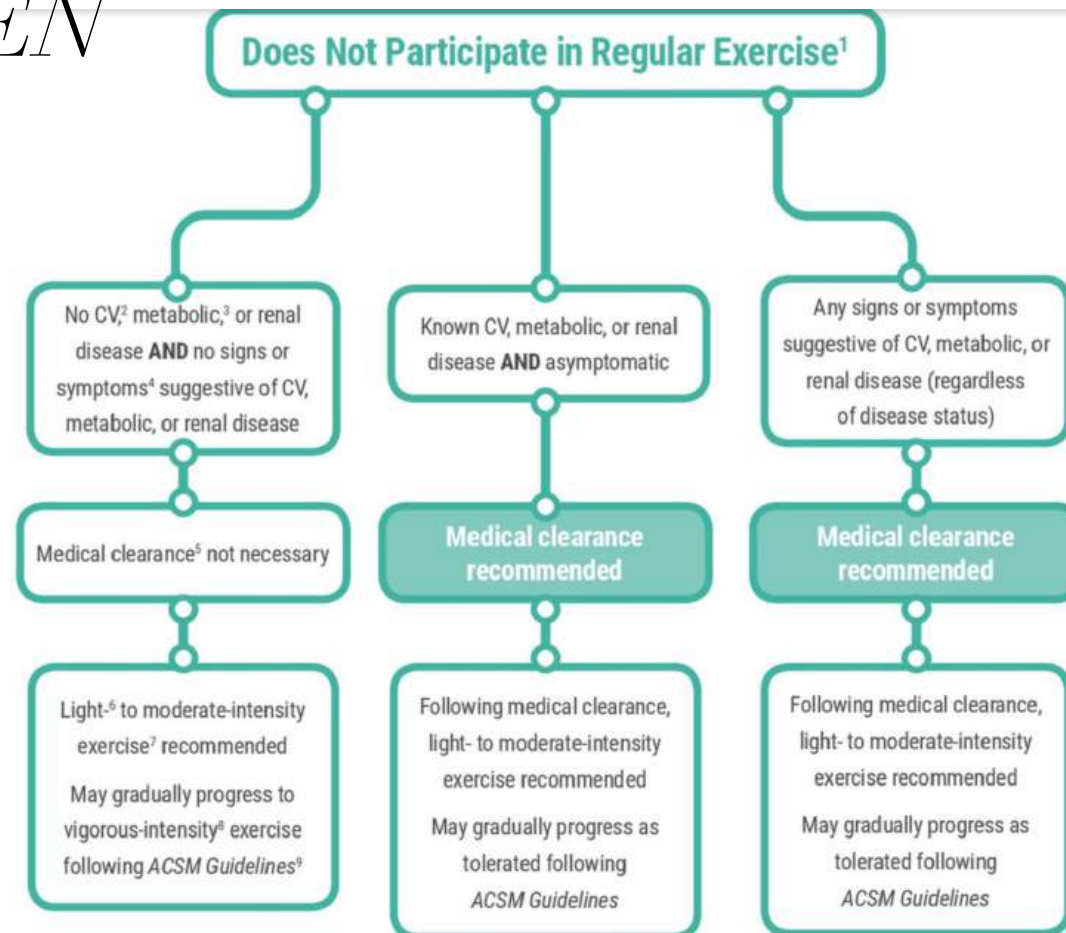


Figure 5-1: The American College of Sports Medicine's preparticipation health-screening algorithm

ACSM PREPARTICIPATION HEALTH SCREEN ALGORITHM

FIGURE 5-1
(continued)

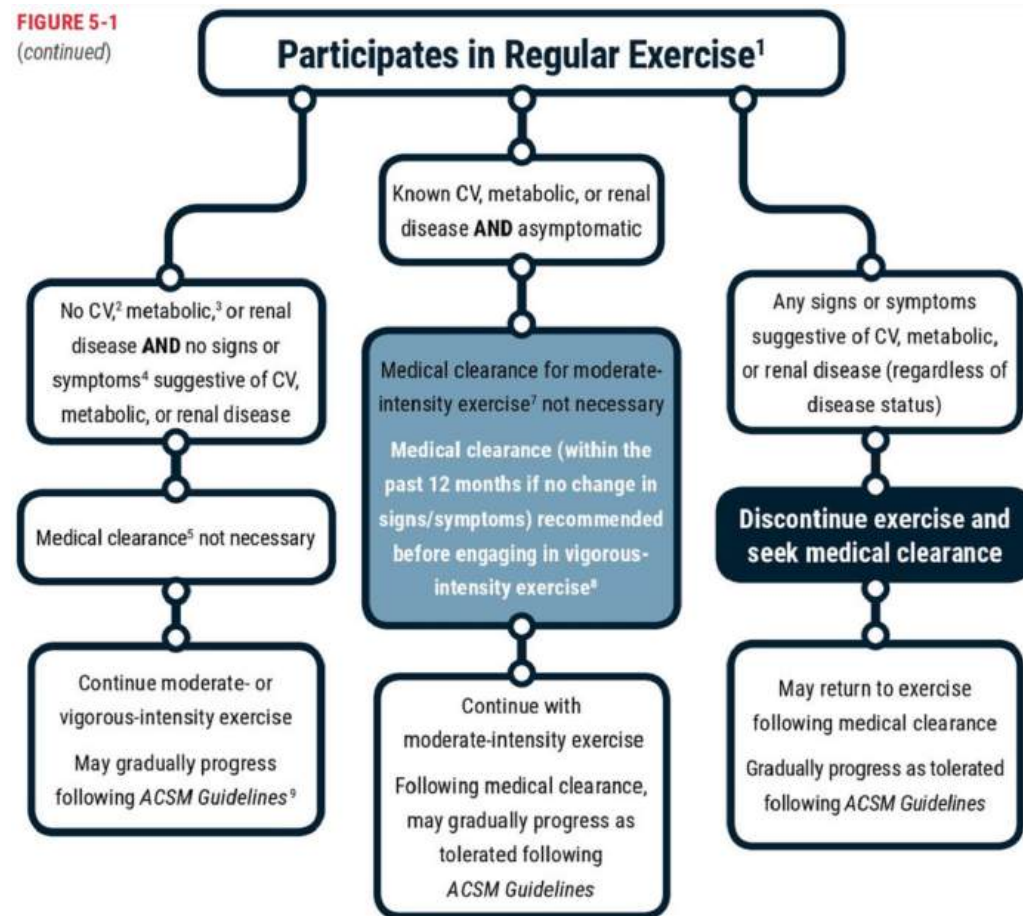


Figure 5-1: The American College of Sports Medicine's preparticipation health-screening algorithm (continued)

SIGNS AND SYMPTOMS SUGGESTIVE OF CVD, METS AND RENAL DISEASE

- Pain or discomfort in the chest (angina), neck, jaw, arms, or other areas that may result from myocardial ischemia
- Dyspnea: shortness of breath at rest or with mild exertion
- Orthopnea: dyspnea in a reclined position
- Paroxysmal nocturnal dyspnea: onset of dyspnea 2-5 h after sleep
- Dizziness or syncope: commonly caused by reduced perfusion to the brain
- Ankle edema: swelling due to excess fluid
- Palpitations or tachycardia: forceful or rapid heartbeat
- Known heart murmur
- Unusual fatigue or shortness of breath



PREPARTICIPATION HEALTH SCREENING

PAR-Q+ is a minimal, yet safe pre-exercise screening measure for low-to-moderate intensity training.

2020 PAR-Q+

The Physical Activity Readiness Questionnaire for Everyone

The health benefits of regular physical activity are clear; more people should engage in physical activity every day of the week. Participating in physical activity is very safe for MOST people. This questionnaire will tell you whether it is necessary for you to seek further advice from your doctor OR a qualified exercise professional before becoming more physically active.

GENERAL HEALTH QUESTIONS

Please read the 7 questions below carefully and answer each one honestly: check YES or NO.	YES	NO
1) Has your doctor ever said that you have a heart condition ☐ OR high blood pressure ☐ ?	☐	☐
2) Do you feel pain in your chest at rest, during your daily activities of living, OR when you do physical activity?	☐	☐
3) Do you lose balance because of dizziness OR have you lost consciousness in the last 12 months? Please answer NO if your dizziness was associated with over-breathing (including during vigorous exercise).	☐	☐
4) Have you ever been diagnosed with another chronic medical condition (other than heart disease or high blood pressure)? PLEASE LIST CONDITION(S) HERE: _____	☐	☐
5) Are you currently taking prescribed medications for a chronic medical condition? PLEASE LIST CONDITION(S) AND MEDICATIONS HERE: _____	☐	☐
6) Do you currently have (or have had within the past 12 months) a bone, joint, or soft tissue (muscle, ligament, or tendon) problem that could be made worse by becoming more physically active? Please answer NO if you had a problem in the past, but it does not limit your current ability to be physically active. PLEASE LIST CONDITION(S) HERE: _____	☐	☐
7) Has your doctor ever said that you should only do medically supervised physical activity?	☐	☐

Figure 5-3: The Physical Activity Readiness Questionnaire for Everyone

Electronic physical activity readiness medical examination (ePARmed-X+)

- The ePARmed-X+ is an easy to follow interactive program that can be used to determine the readiness for increased physical activity participation or a fitness appraisal.
- Who is it for:
 - Clients with concerning medical conditions may be referred to this by a qualified exercise professional or another healthcare professional
 - Client who has a positive response to PAR-Q+

OTHER USEFUL FORMS AND TOOLS

Date _____

Dear Doctor:

Your patient, _____, wishes to start a personalized training program. The activity will involve the following:

(type, frequency, duration, and intensity of activities)

If your patient is taking medications that will affect his or her exercise capacity or heart-rate response to exercise, please indicate the manner of the effect (raises or lowers exercise capacity or heart-rate response):

Type of medication(s) _____

Effect(s) _____

Please identify any recommendations or restrictions that are appropriate for your patient in this exercise program:

Thank you.
Sincerely,

Fred Fitness
Personalized Gym
Address _____
Phone _____

_____ has my approval to begin an exercise program with the recommendations or restrictions stated above.

Signed _____ Date _____

Phone _____

- Risk stratification criteria from AACVPR for working with clients with known CVD in clinical settings (Pg 149)
- Lifestyle and health-history questionnaire (Fig. 5-4)
- Medical release form (Fig. 5-5)
- Informed consent/Assumption of risk (Fig. 16-2)
- Agreement to participate (Fig. 16-1)

Example of form that can be used for clients

ACSM RISK STRATIFICATION

Risk Factors	Defining Criteria	Points
Age	Men ≥ 45 , Women ≥ 55	+1
Family History	Myocardial infarction, coronary revascularization, or sudden death before 55-year-old in first-degree male relative, or before 65-year-old in first-degree female relative	+1
Cigarette Smoking	Current cigarette smoker or those who quit within the previous six months, or exposure to environmental tobacco smoke	+1
Sedentary Lifestyle	Not participating in at least 30 minutes of moderate-intensity physical activity on at least three days/week for at least three months	+1
Obesity	BMI ≥ 30 kg/m ² or waist girth > 100 cm for men and > 90 cm for women	+1
Hypertension	Systolic blood pressure ≥ 140 mmHg and/or diastolic blood pressure ≥ 90 mmHg, confirmed by measurements on at least two separate occasions, or currently on antihypertensive medication	+1
Dyslipidemia	LDL cholesterol ≥ 130 mg/dL or HDL < 40 mg/dL or on lipid-lowering medication	+1
Prediabetes	Fasting plasma glucose ≥ 100 mg/dL but ≤ 125 mg/dL or oral glucose tolerance test value of ≥ 140 mg/dL but ≤ 199 mg/dL, confirmed by measurements on at least two separate occasions	+1
Negative Risk Factor	HDL cholesterol ≥ 60 mg/dL	-1

EFFECT OF MEDICATIONS ON HEART RATE RESPONSE

Anti-hypertensives primarily affect one of four different sites:



- Heart: to reduce its force of contraction.

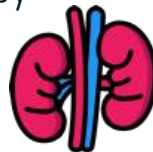


- Peripheral blood vessels: to open or dilate them to allow more room for the blood.



- Brain: to reduce the sympathetic nerve outflow.

- Kidneys: to reduce blood volume by excreting more fluid.



The most common antihypertensives:

- Beta blockers.
- Calcium channel blockers.
- Angiotensin-converting enzyme (ACE) inhibitors.
- Angiotensin-II receptor antagonists.
- Diuretics.



EFFECT OF SELECT SUBSTANCES ON HR RESPONSE

Medications	Resting HR	Exercise HR	Exercise Capacity	Comments
Beta blockers	↓	↓	↓ $\dot{V}O_2$ max with acute and with chronic administration ↑	Dose-related response
Angiotensin II receptor blockers (ARBs) and calcium channel blockers (CCBs)	↓ or ↔	↓ or ↔	↔	
Other antihypertensives*	↑, ↔, or ↓	↑, ↔, or ↓	Usually ↔	Many antihypertensive medications are used. Some may decrease, a few may increase, and others do not affect HR. Some exhibit variable and dose-related responses.
Antihistamines	↑	↔	↔ performance and endurance	
Diuretics	↔	↔	↔	Can cause water/electrolyte imbalance and dehydration

Table 5-1: Effects of Select Substances on Heart-rate Response



CHAPTER 7

RESTING ASSESSMENTS AND ANTHROPOMETRIC MEASUREMENTS

PART II. RESTING ASSESSMENTS



Learning Objectives:

- Explain the need for proper sequencing of assessments
- Describe how to measure heart rate and blood pressure at rest and the conditions under which they should be measured
- Identify the various methods to estimate body composition and explain the advantages and disadvantages of each
- Explain the practical reasons for using body mass index, as well as limitations for estimating overweight and obesity
- Identify anatomical landmarks and assessment protocols associated with various circumference measurements

BUILD SELF-EFFICACY BY CHOOSING THE RIGHT ASSESSMENTS

Identify areas of
health / injury risk



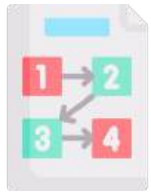
Collect baseline data

Educate clients

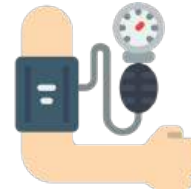


Motivate clients

WHAT YOU WILL LEARN



Sequencing
assessments

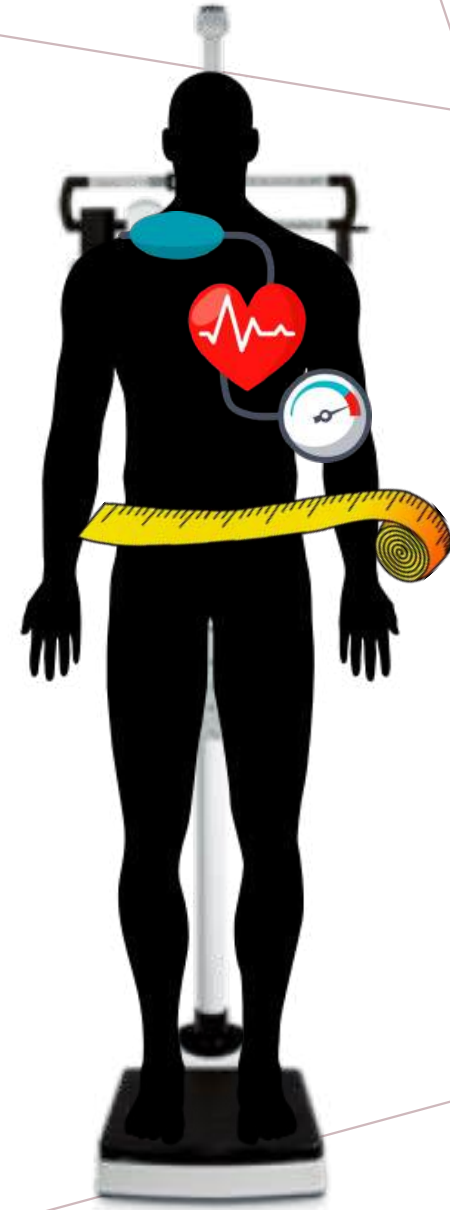
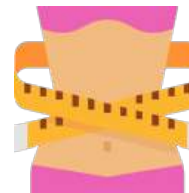


Cardiovascular assessments at rest

- Heart Rate
- Blood Pressure

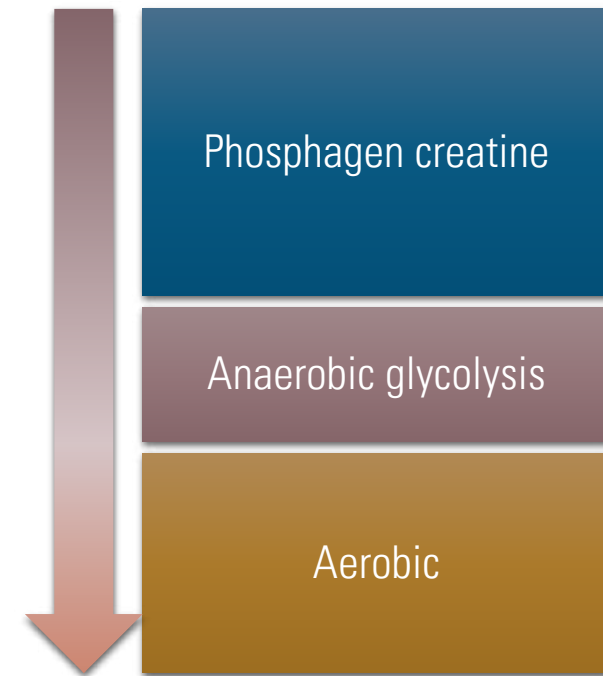
Anthropometric measurements

- Height, Weight and BMI
- Body Composition
- Girth Measurements



ASSESSMENTS – SEQUENCING

1. Resting tests: HR, BP, height, weight, body composition
2. Non-fatiguing tests: postural, flexibility, balance, vertical jump
3. Agility: e.g. pro-agility, T-test
4. Maximal power and strength: e.g. power clean, 3 RM, 1RM
5. Sprint tests: e.g. 40-yard sprint
6. Local muscular endurance: e.g. YMCA bench press, partial curl-up
7. Anaerobic capacity: e.g. 300-yard shuttle
8. Submaximal aerobic capacity: e.g. step test, Rockport walking test, YMCA cycle ergometer test, 1.5-mile run, 12-minute run/walk



Most appropriate to perform aerobic test on a separate day. However, if all assessments are performed on the same day, maximum aerobic tests should be performed last, after a minimum of an hour-long rest recovery period.

ASSESSMENT – HEART RATE

Assess HR via palpation, auscultation, or HR monitors.

- Radial artery – the ventral aspect of the wrist medial to the radius
- Carotid artery – located in the neck, lateral to the trachea
 - Pressing too hard $\Rightarrow$ slowed HR
- Brachial artery – medial to the humerus



Figure 7-1: Taking the pulse of the radial artery



Figure 7-2: Taking the pulse of the carotid artery

HEART RATE

<60		70-72		>100
Bradycardia		Avg		Tachycardia

Heart rate is a valid indicator of work intensity or stress on the body:



Males (60-70 bpm)



Females (72-80 bpm)

- At rest, measure for 30 to 60 sec
- During exercise, measure for 15 sec and multiply by 4

HEART RATE

Factors Affecting Heart Rate Assessment

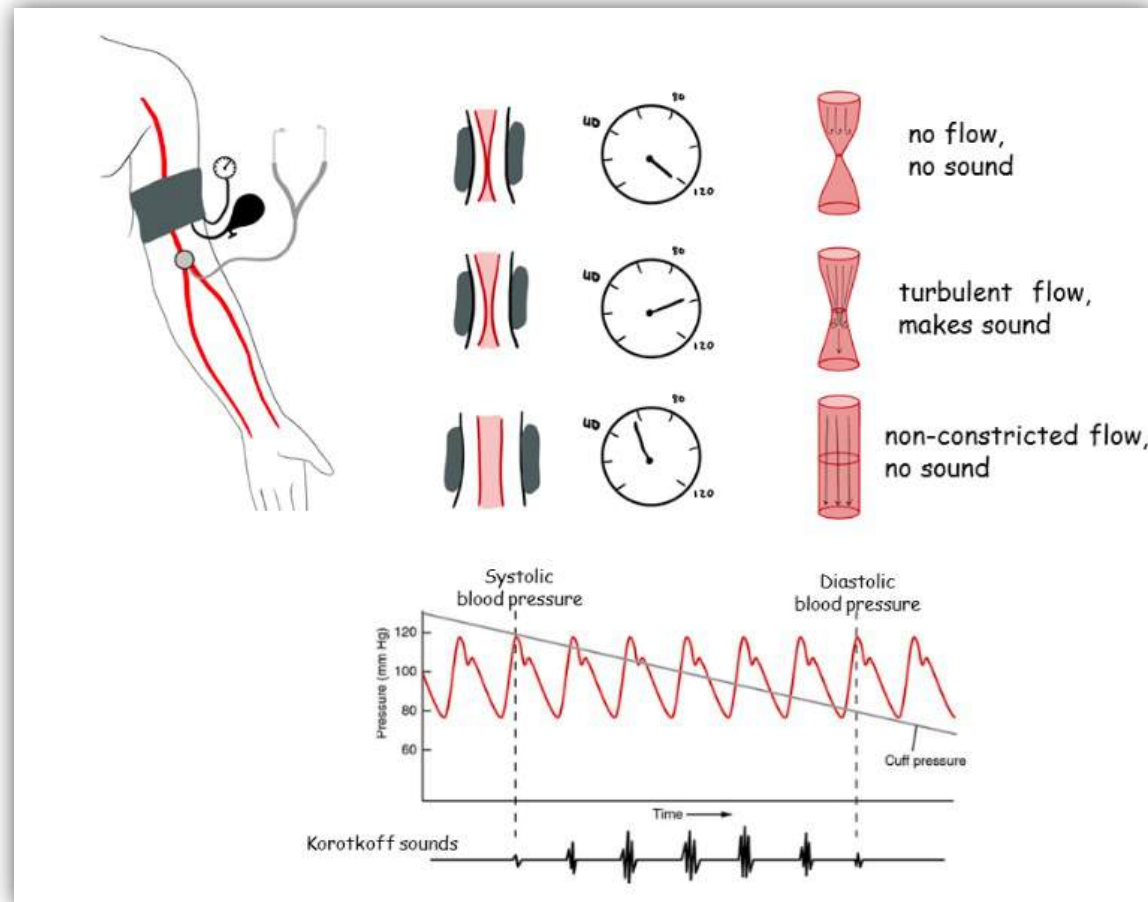
- Smoking and tobacco products (↑ resting HR; ↑ or -- exercise HR)
- Caffeine (↑ or resting and exercise HR)
- Environmental temperatures (hot temperatures: ↑ RHR and exercise HR in; cold temperatures: variable)
- Altitude (↑ HR at altitudes greater than approximately 4,000 feet or 1,200m)
- Stress (↑ resting and exercise HR)
- Body position (↓ HR when supine, ↑ HR from supine to seated position or standing position)
- Time of day (↓ HR in the morning, ↑ or -- during afternoon or evening)
- Medications (variable)

Note: ↑ = increase, ↓ = decrease, -- = no significant change



BLOOD PRESSURE

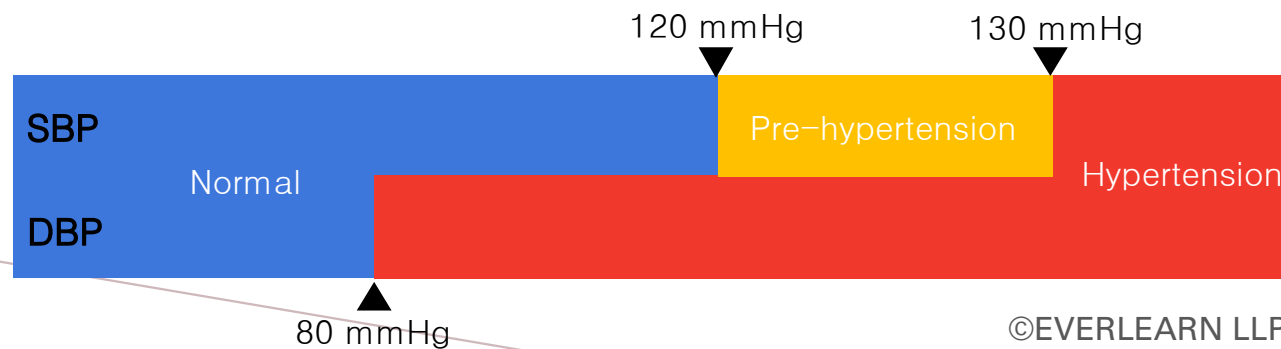
Blood Pressure is measured indirectly by listening to the Korotkoff sounds



BLOOD PRESSURE

Category	SBP		DBP
Normal	<120 mmHg	and	<80 mmHg
Elevated	120–129 mmHg	and	<80 mmHg
Hypertension			
Stage 1	130–139 mmHg	or	80–89 mmHg
Stage 2	≥140 mmHg	or	≥90 mmHg

Table 7-2: Categories of Blood Pressures in Adults



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BLOOD PRESSURE

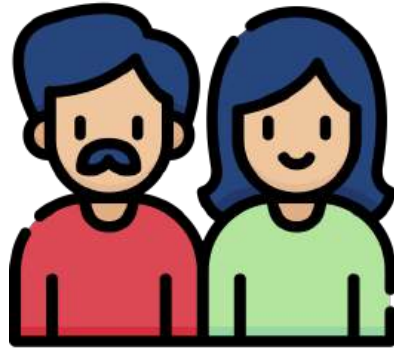
Factors Affecting Blood Pressure Assessment

- Smoking and tobacco products (↑ resting and exercise)
- Caffeine (variable)
- Stress (↑ resting and exercise)
- Body position (↓ when supine, ↑ from supine to seated position or standing position)
- Time of day (↓ first thing in the morning, ↑ or -- during afternoon or evening hours)
- Medications (variable)

Note: ↑ = increase, ↓ = decrease, -- = no significant change



BLOOD PRESSURE



For individuals 40-70 years old, each 20 mmHg increase in SBP or 10 mmHg increase in DBP above the normal doubles the risk of CVD



A PT discovering an abnormal BP reading should recommend that the client visit their personal physician



A difference of 15 mmHg or more between arms increases the risk of PVD and cerebral vascular disease

ANTHROPOMETRIC MEASUREMENTS



BODY MASS INDEX (BMI)

BMI	Classification
<18.5	Underweight
18.5-25	Normal
25-30	Overweight
30-35	Obesity Class I
35-40	Obesity Class II
≥ 40	Obesity Class III

Formula:

$$\text{BMI} = \text{Weight (kg)} / \text{Height}^2 (\text{m}^2)$$

Table 7-6: Classification of Overweight and Obesity by Body Mass Index (BMI)

CIRCUMFERENCE MEASUREMENTS

[PG 239-240]

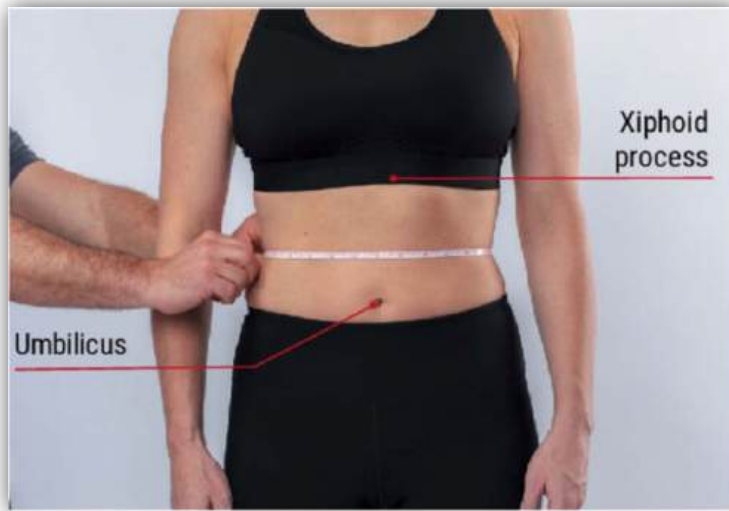


Figure 7-8: Waist circumference

Waist circumference: the narrowest part of the torso (above umbilicus and below the xiphoid process)

Hip circumference: the maximal circumference of the buttocks



Figure 7-7: Hip circumference

CIRCUMFERENCE MEASUREMENTS - WAIST

For every inch increase in waist circumference in men,

- BP increases by 10%
- Blood cholesterol level increases by 8%
- Triglycerides increase by 18%
- HDL decreases by 15%
- MetS risk increases by 18%

Waist Circumference		
Risk Category	Male	Female
Very low	<80 cm	<70 cm
Low	80-99 cm	70-89 cm
High	100-120 cm	90-110 cm
Very high	>120 cm	>110 cm

Table 7-7: Criteria for Waist Circumference in Adults

CIRCUMFERENCE MEASUREMENTS

– WAIST-TO-HIP RATIO

		RISK			
	Age	Low	Moderate	High	Very High
Men	20–29	<0.83	0.83–0.88	0.89–0.94	>0.94
	30–39	<0.84	0.84–0.91	0.92–0.96	>0.96
	40–49	<0.88	0.88–0.95	0.96–1.00	>1.00
	50–59	<0.90	0.90–0.96	0.97–1.02	>1.02
	60–69	<0.91	0.91–0.98	0.99–1.03	>1.03
Women	20–29	<0.71	0.71–0.77	0.78–0.82	>0.82
	30–39	<0.72	0.72–0.78	0.79–0.84	>0.84
	40–49	<0.73	0.73–0.79	0.80–0.87	>0.87
	50–59	<0.74	0.74–0.81	0.82–0.88	>0.88
	60–69	<0.76	0.76–0.83	0.84–0.90	>0.90

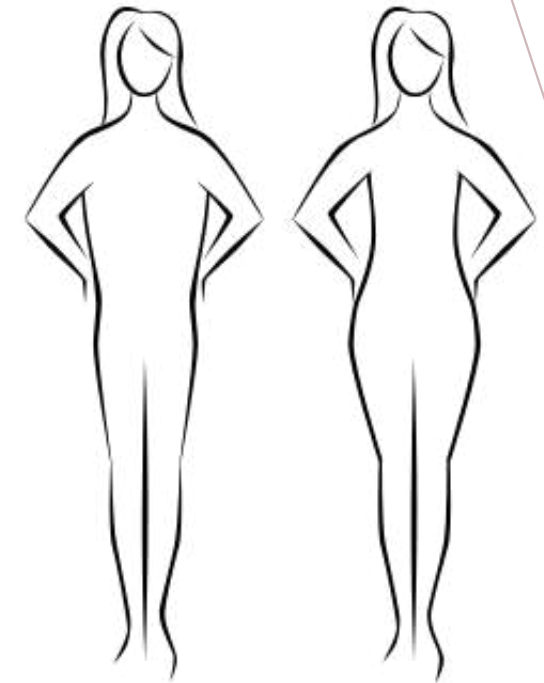
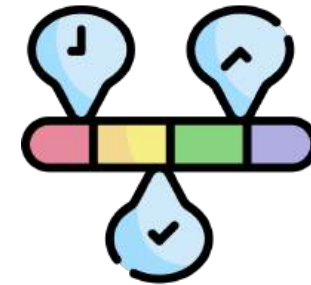
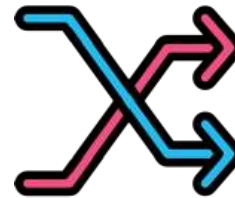


Table 7-8: Waist-to-Hip Circumference Ratio Norms for Men and Women

ACE MOVER METHOD



Health Outcome	Expected Change	Timeline
Body fat (%)	↓ 2.7-5.9	2-3 months
Waist circumference (cm)	↓ 1.9-3.3	2-3 months
Body weight	↓ 1-2 lb/week	

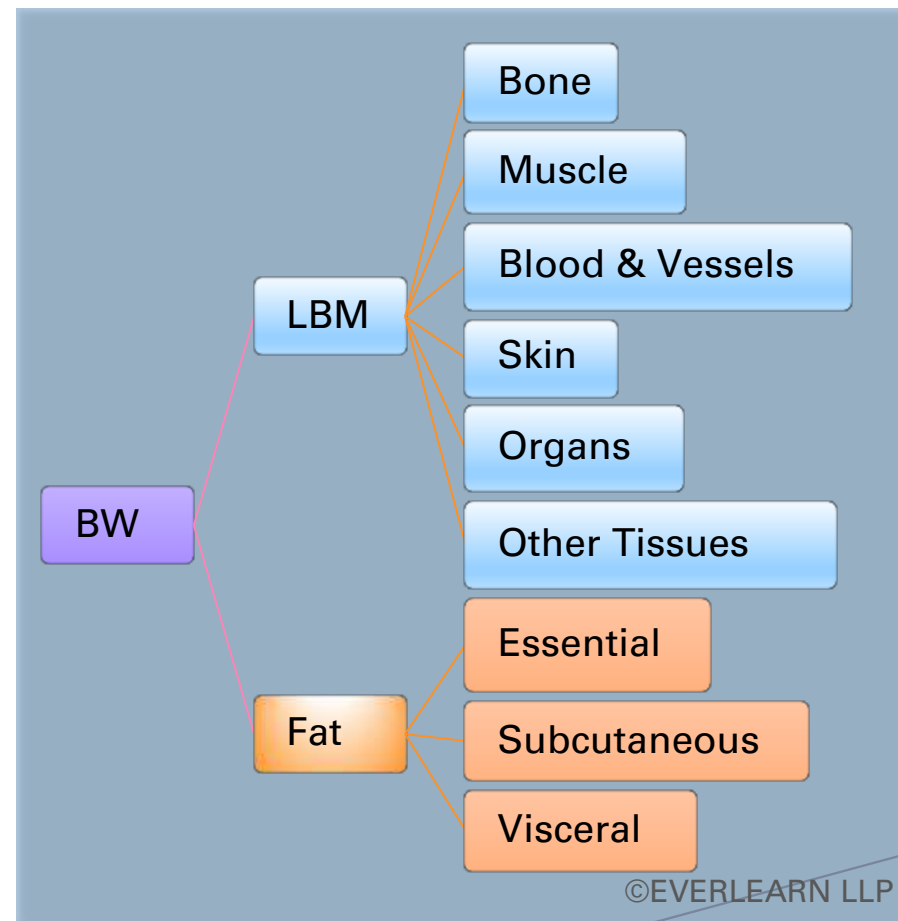
The Expected Change and Timeline for Key Anthropometric and Body-Composition Measures

More information on ACE Mover Method can be found at: <https://www.acefitness.org/about-ace/ace-mover-method/#:~:text=The%20ACE%20Mover%20Method%20is,and%20trust%2C%20communication%20and%20collaboration.>

BODY COMPOSITION

[PG 227 & 232]

	Fat%	
Classification	Male	Female
Essential Fat	2-5	10-13
Average	18-24	25-31
Obese	>24	>31



BODY COMPOSITION

[PG 228-229]

Method	Description
Bioelectrical impedance analysis (BIA)*	Measures electrical signals as it passes through fat, lean mass, and water in the body. BIA accuracy is based primarily on the sophistication of the machine and the validity of the prediction algorithms. Optimal hydration is necessary for accurate results.
Dual-energy x-ray absorptiometry (DEXA) *	Among the most accurate and precise methods. It is a whole-body scanning system that delivers a low-dose x-ray that reads bone and soft-tissue mass. It can identify regional body-fat distribution
Magnetic resonance Imaging (MRI)* Found in hospitals and diagnostic centers	Uses magnetic fields to assess how much fat a person has and where it is deposited. Since MRIs are located in clinical settings, using an MRI solely for calculation of body fat is not practical.
Total body electrical conductivity (TOBEC) Found in clinical and research settings	TOBEC uses an electromagnetic force field to assess relative body fat. Much like the MRI, it is impractical and too expensive for the fitness setting.

Table 7-3: Body-composition Assessments

BODY COMPOSITION (CON'T)

[PG 228-229]

Method	Description
Hydrostatic weighing (i.e. underwater weighing)	Measures the amount of water displacement when completely submerged, thereby indirectly measuring body fat via body density. Generally not practical in a fitness center setting due to size of the apparatus and the complexity of the technique required for accurate measurement. The assessment must then be repeated to ensure reliability.
Air displacement plethysmography (ADP) Example: Bod Pod	The Bod Pod is an egg-shaped chamber that measures the amount of air displaced when a person sits in the machine. ADP has a high accuracy rate but the equipment is expensive
Near-infrared interactance (NIR) Example: Futrex	Uses a fiber optic connected to a digital analyser that indirectly measures tissue composition (fat and water). Typical assessment site is the biceps. Calculations take into consideration of height, weight, frame size, and level of activity. It is relatively inexpensive and fast, but generally not as accurate as other techniques
Skinfold measurement Very commonly used in fitness settings	Skinfold calipers are used to "pinch" a fold of skin and fat. Several sites on the body are typically measured. The measurements are plugged into an equation that calculates body-fat percentage.

Table 7-3: Body-composition Assessments

$$BMI = \frac{250}{71^2} \times 703 = \frac{250}{5,041} = 34.9 \text{ kg/m}^2$$

ACE → MOVER™ METHOD

CHAPTER 8
CARDIORESPIRATORY TRAINING:
PHYSIOLOGY, ASSESSMENTS AND
PROGRAMMING

PART III. CARDIORESPIRATORY TRAINING: PHYSIOLOGY AND ASSESSMENTS

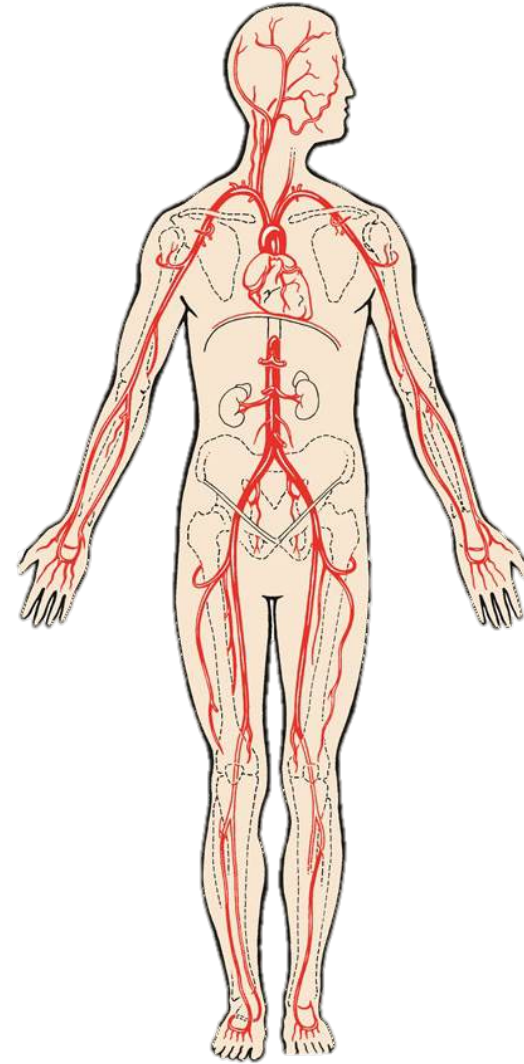
Learning Objectives:

- Identify the acute and chronic physiological responses to cardiorespiratory training
- Recognize key environmental considerations for cardiorespiratory training
- Explain the procedures for ventilatory threshold assessment
- Evaluate the results of cardiorespiratory fitness assessments and apply them to cardiorespiratory programming

CARDIOVASCULAR SYSTEM

The cardiovascular system, also called the circulatory system, is composed of the heart, blood vessels, and blood.

Blood is the fluid component that transports necessary substances throughout the body.



BLOOD FLOW THROUGH THE HEART

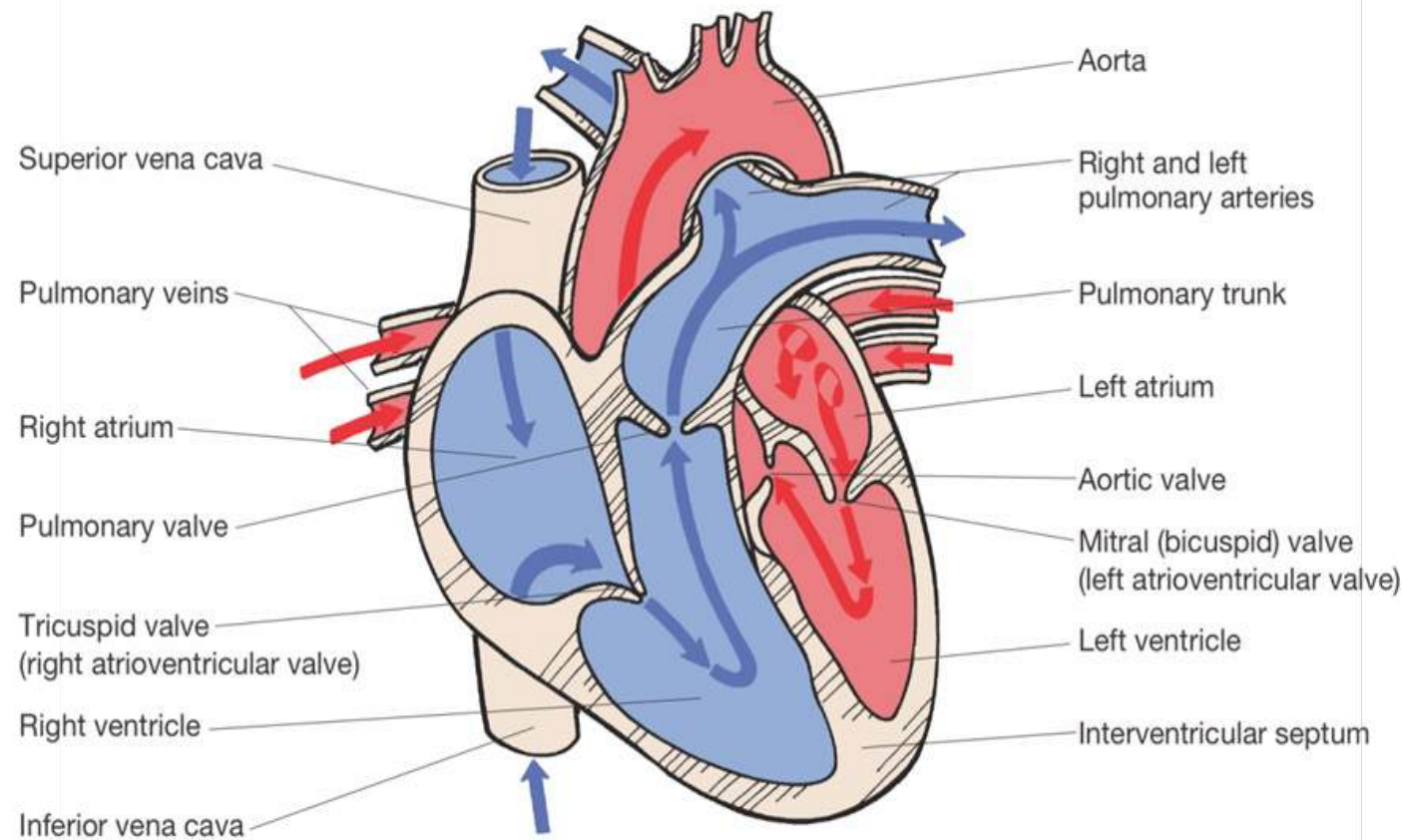
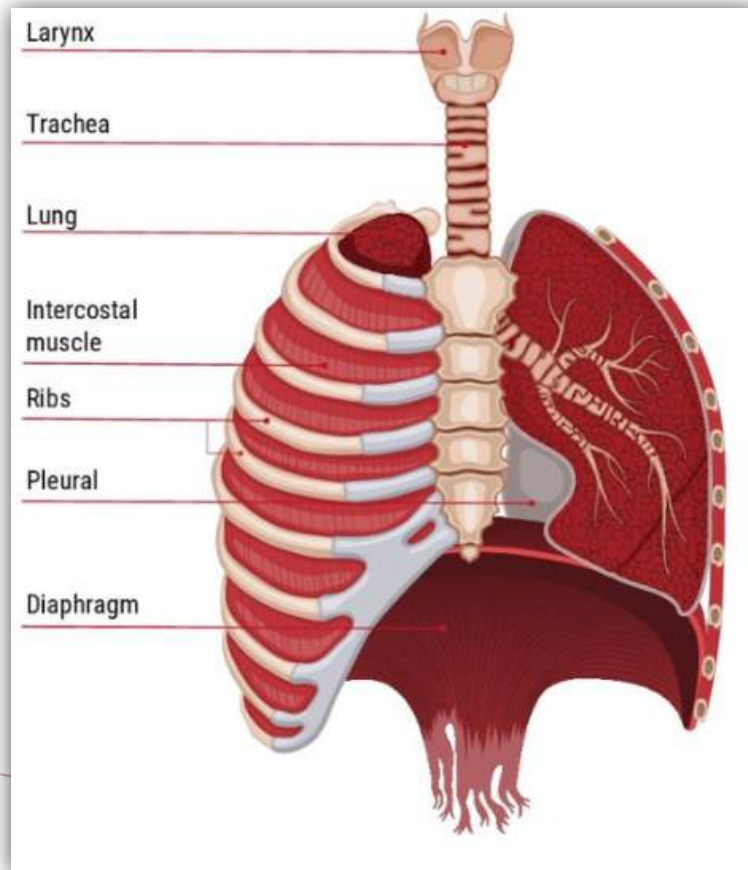


Figure 8-2: Structure of the heart and flow of blood within it

FUNCTIONS OF RESPIRATORY SYSTEM

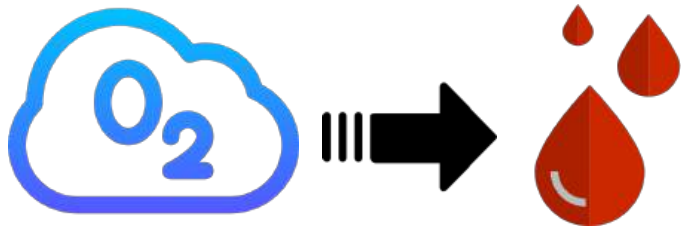


- Replacing oxygen and removing carbon dioxide
- Vocalization
- Regulation of the acid-base balance during exercise

Figure 8-4: Diaphragm and intercoastal muscles

PHYSIOLOGY OF THE CARDIORESPIRATORY SYSTEM

The cardiorespiratory system is responsible for three basic processes:



Get oxygen into the blood (coupled with oxygen-carrying capacity of the circulatory system)



Distribute blood flow and deliver oxygen to the active tissues (a function of cardiac output)



Extract O₂ from the blood to produce ATP

O₂-CARRYING

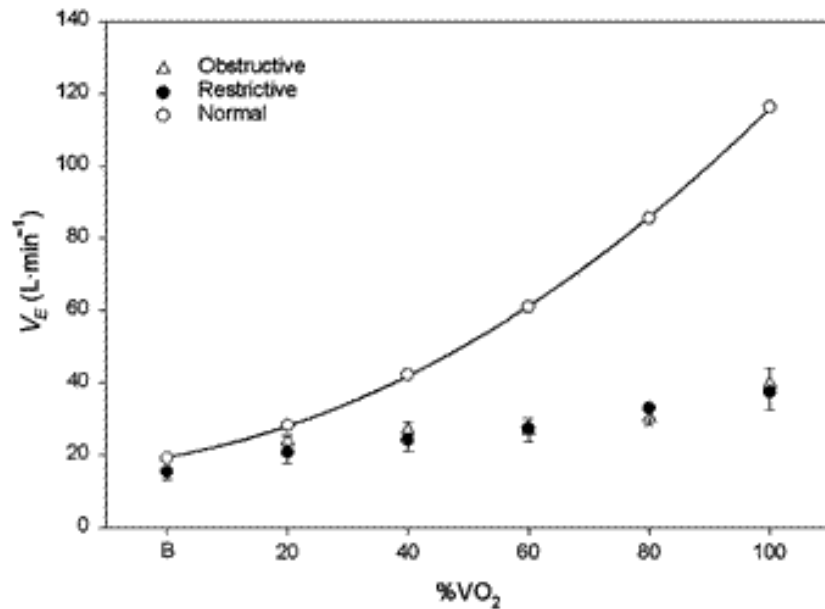


Figure 1. Relationship between minute ventilation (V_E) and exercise intensity relative to maximum oxygen (% VO_2 max) subjects with obstructive (Δ) and restrictive ($\bullet$) ventilatory abnormalities. Reference values of normal subjects (based on unpublished laboratory data) are shown with their regression (--).

- Pulmonary ventilation (V_E): volume of air exchanged per minute
- V_E = breathing frequency · tidal volume
- Ventilation increases linearly with VO_2 up to about 50% VO_{2max} , after which the increase is proportionately greater than the increase in work rate
- V_E is perhaps regulated more by the requirements for CO_2 removal than by O_2 consumption
- Ventilation is normally not a limiting factor to aerobic capacity in healthy individuals
- Individuals with emphysema or asthma can have reduced respiratory capacity

Resource from: <https://www.rehab.research.va.gov/jour/03/40/5/Nield.html>

O_2 -CARRYING — O_2 -DELIVERY

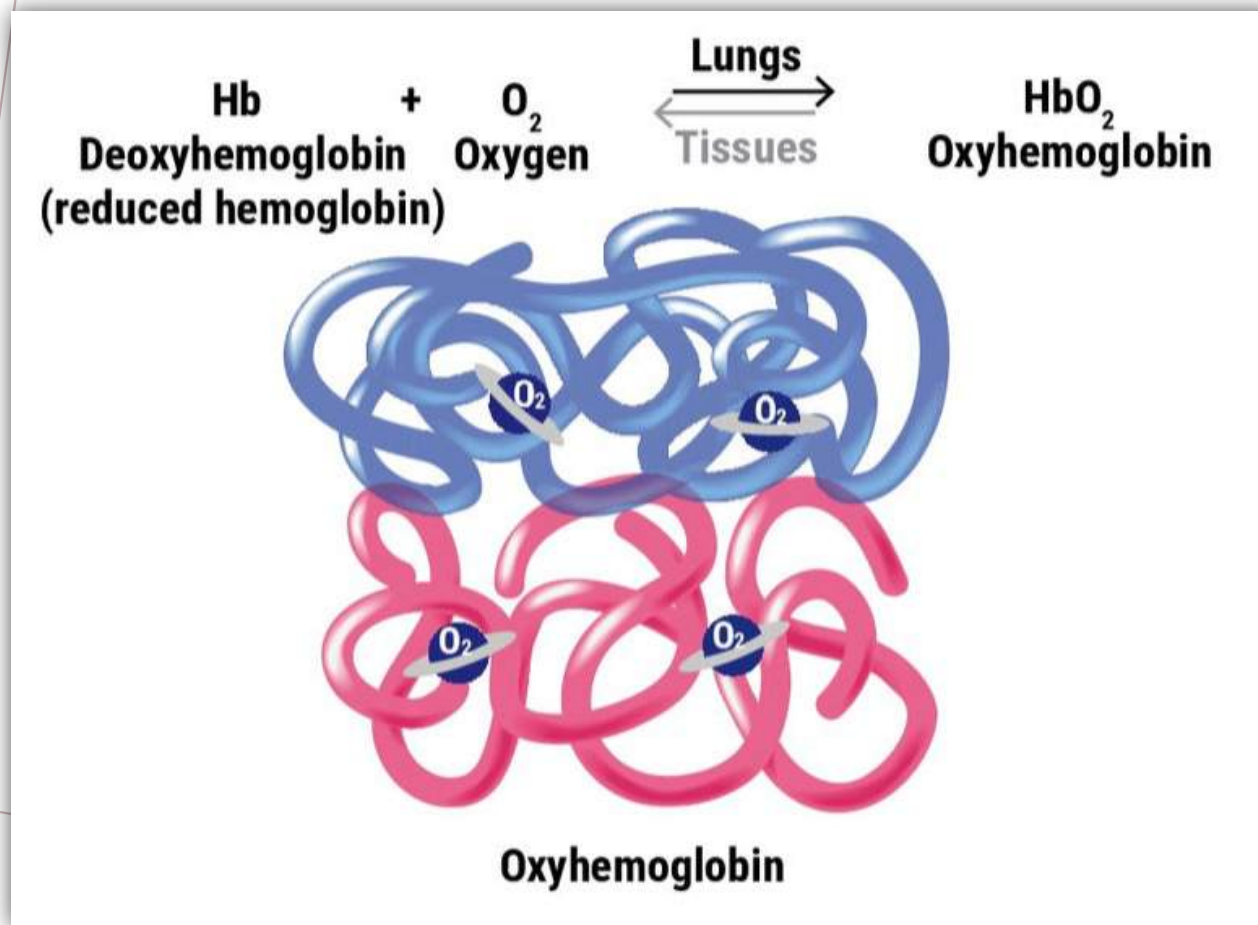
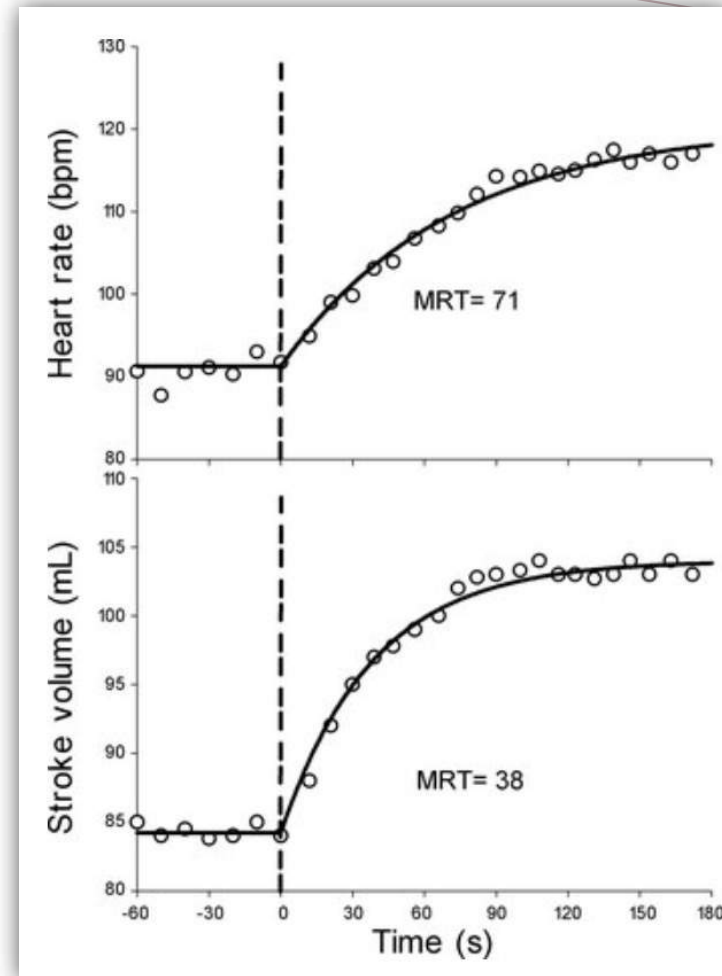


Figure 8-5: Molecules of oxygen (O_2) bind to the protein hemoglobin in red blood cells

O₂-DELIVERY

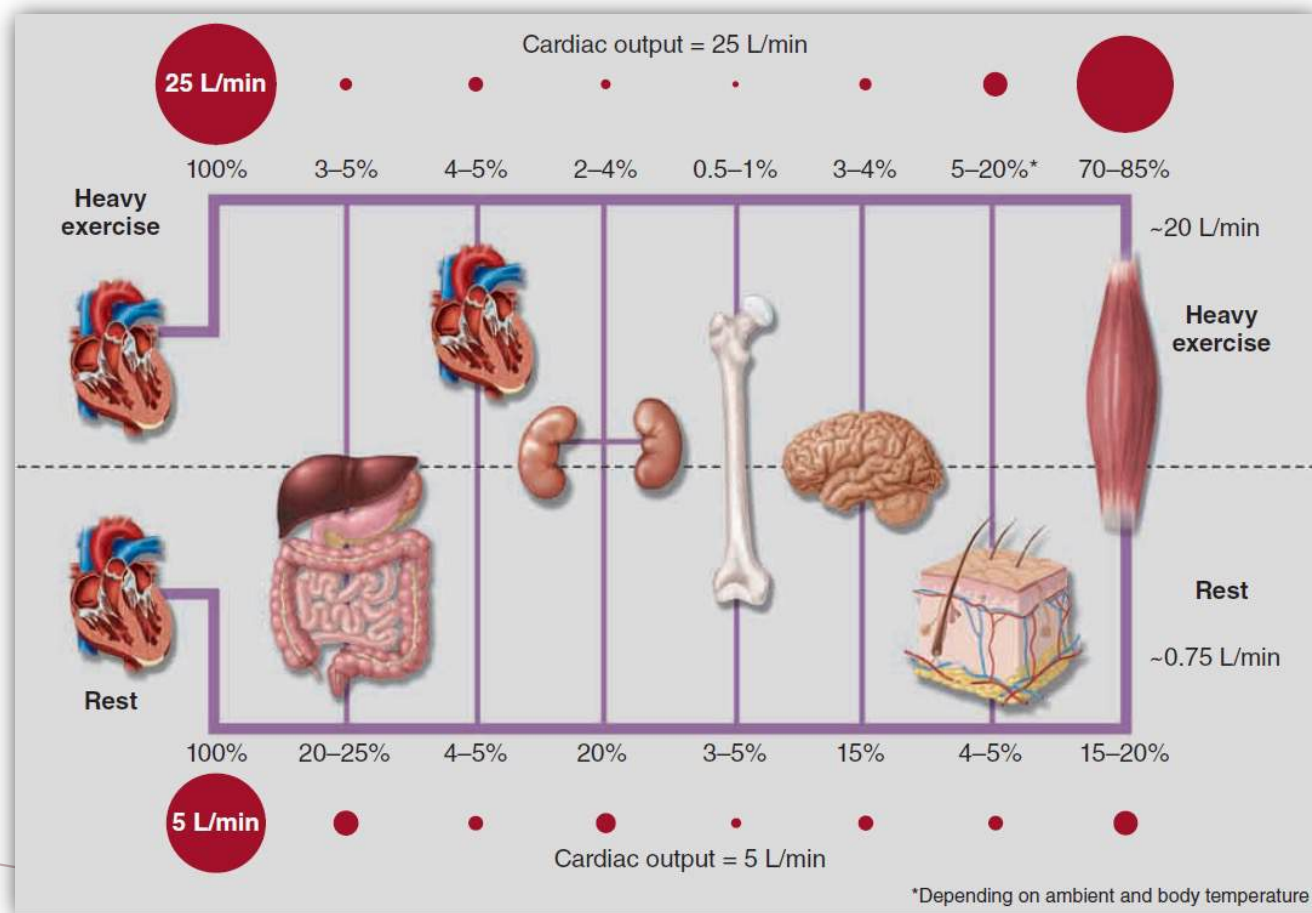
- Cardiac output (*Q*, L/min):
 $HR(\text{bpm}) \times SV(\text{L/beat})$
- *Q* in healthy adults increases linearly with increased work rate
- Maximum values of *Q* depend on many factors including age, posture, body size, presence of cardiovascular disease, and the level of physical conditioning



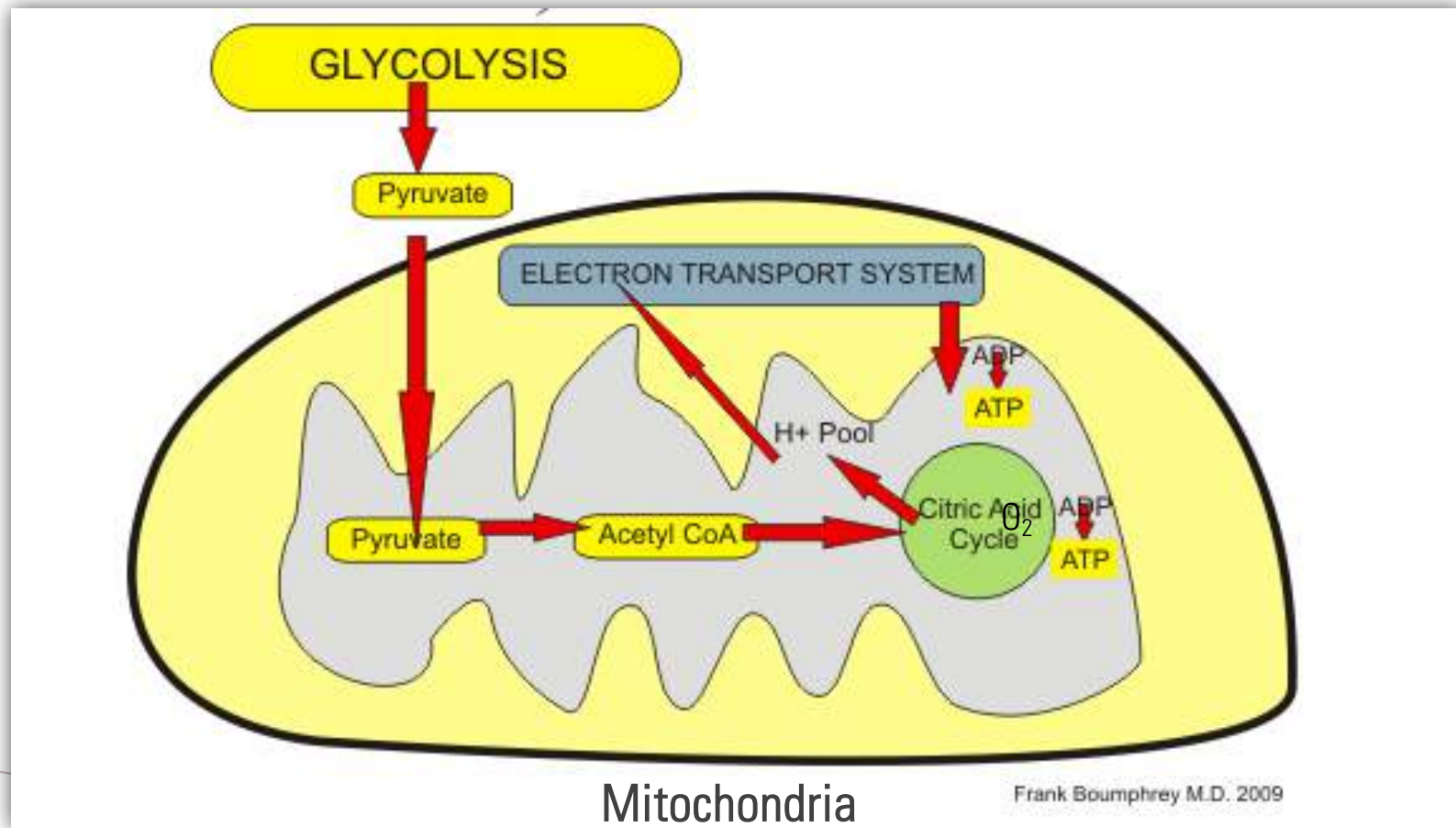
Extracted from G. R. Chiappa *et al.*, *Journal of Applied Physiology*, vol. 104, no. 5, pp. 1341–1350, May 2008, doi: [10.1152/jappphysiol.01364.2007](https://doi.org/10.1152/jappphysiol.01364.2007).

O₂-DELIVERY

[PG 258-259]



O_2 EXTRACTION

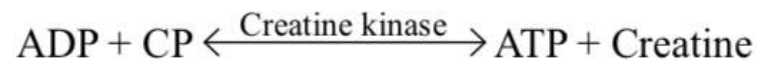


O_2 UTILIZATION

Bioenergy Systems

ANAEROBIC

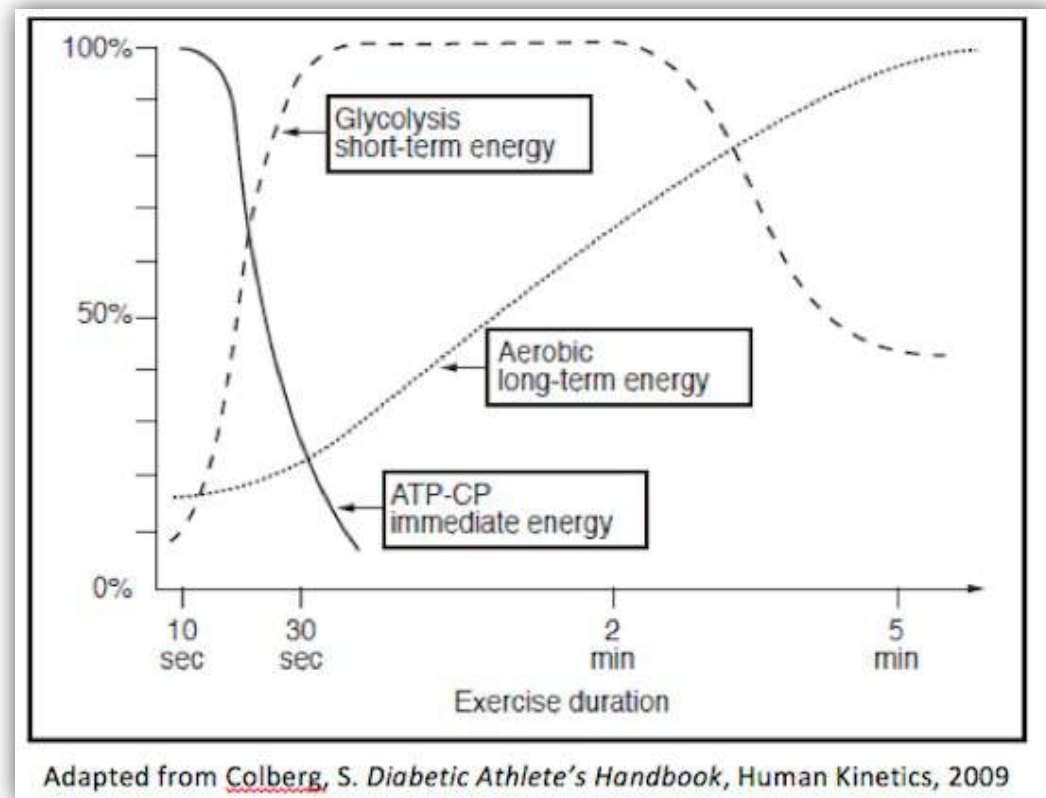
- ATP-CP



- Anaerobic glycolysis
 - Glycogen $\rightarrow$ lactate

AEROBIC

- Glycogen, fat $\rightarrow$ CO_2



CARDIORESPIRATORY CAPACITY

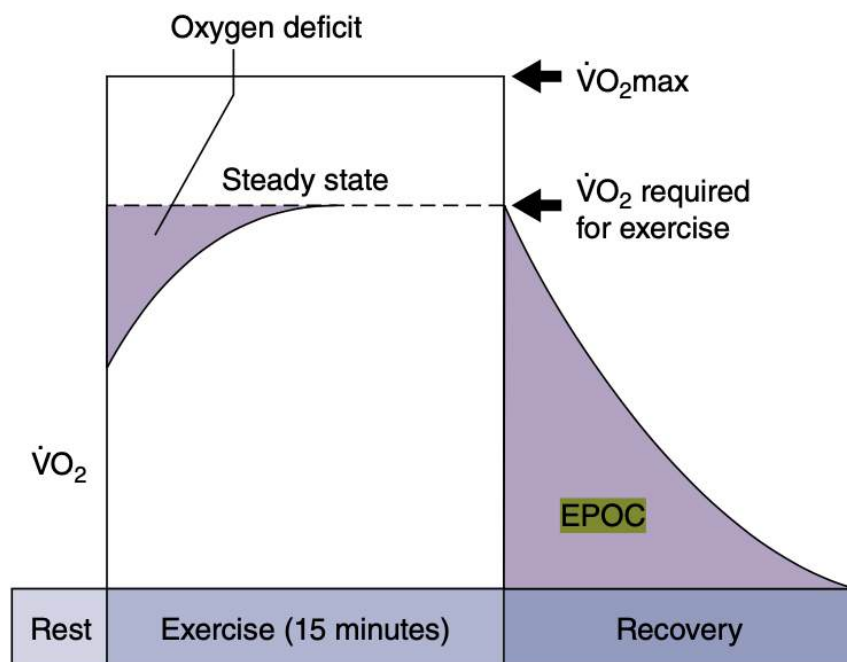
Maximal oxygen uptake (VO_2max , mL/kg·min)

- The maximum rate of oxygen consumption measured during a graded exercise test
- Provides a quantitative value of endurance fitness

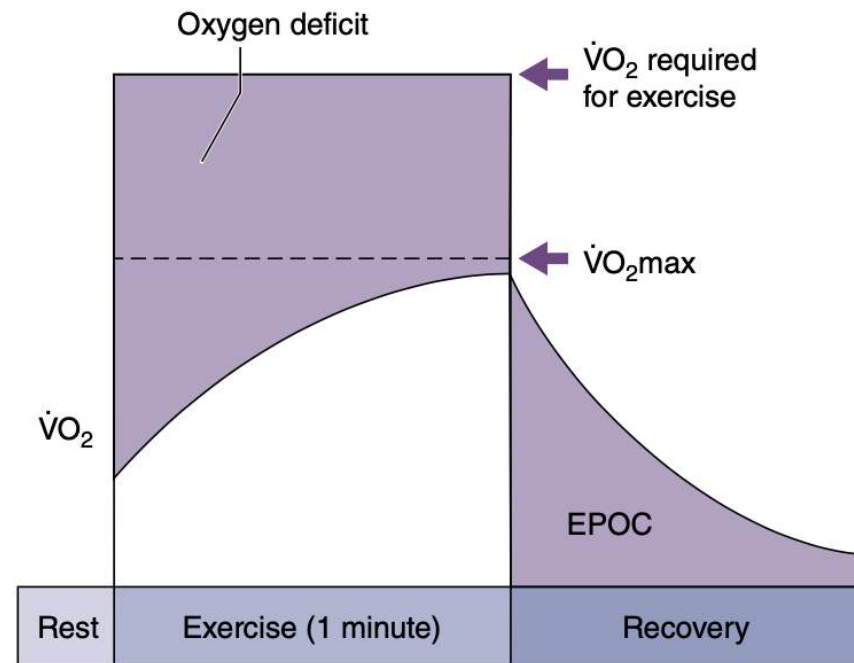


ACUTE RESPONSE TO CARDIORESPIRATORY EXERCISE

OXYGEN DEFICIT AND EPOC



Low intensity, steady state

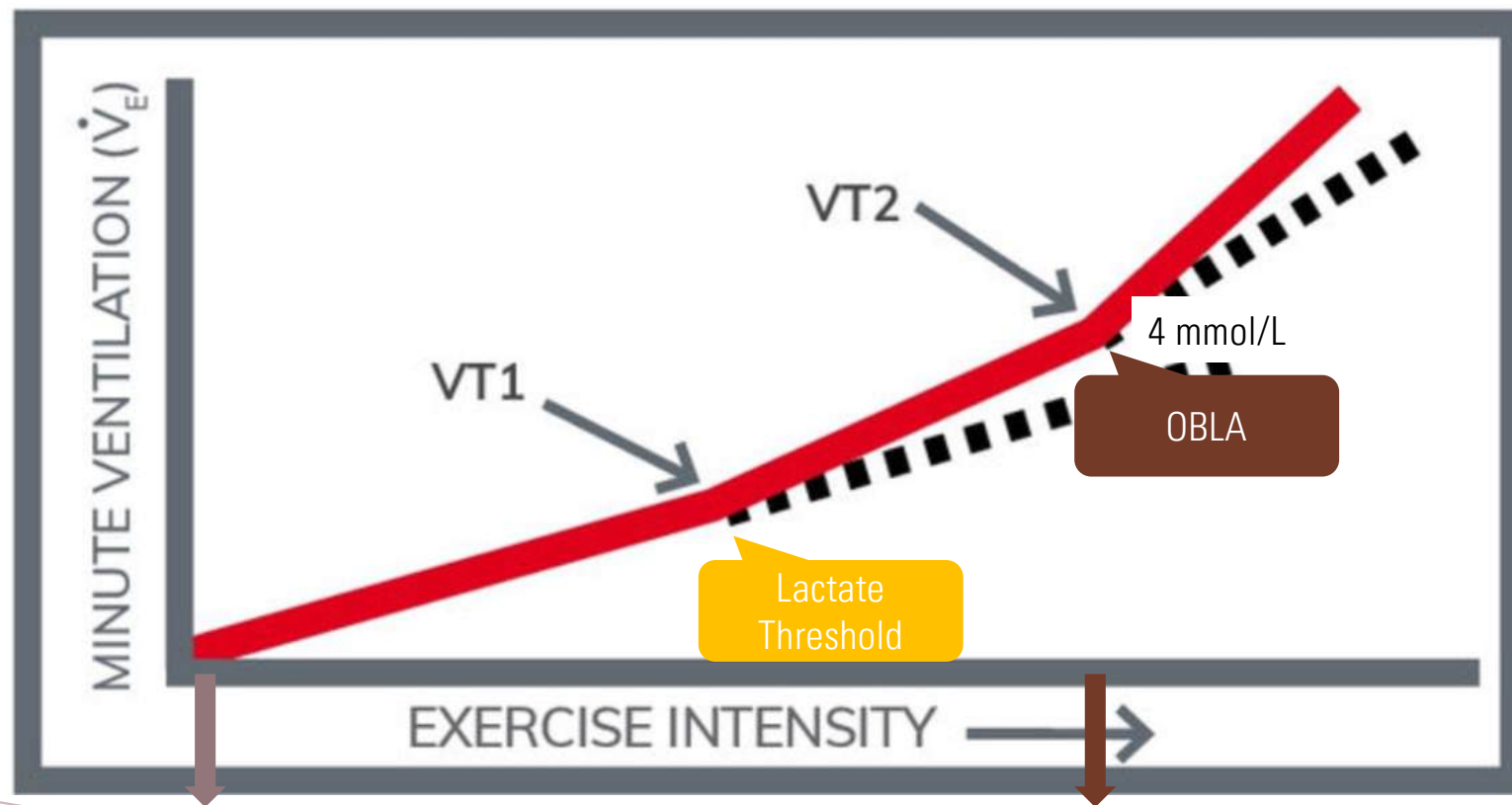


High intensity, non-steady state

Resource from: G. Haff, N. T. Triplett, and National Strength & Conditioning Association (U.S.), Eds., *Essentials of strength training and conditioning*, Fourth edition. Champaign, IL: Human Kinetics, 2016.

VENTILATORY RESPONSE

[PG 260 & 297]



12-15 breaths/min
Tidal V 0.4-1.0 L

35-45 breaths/min
Tidal V > 3 L



RESPONSE TO ENVIRONMENTAL FACTORS

EXERCISING IN THE HEAT

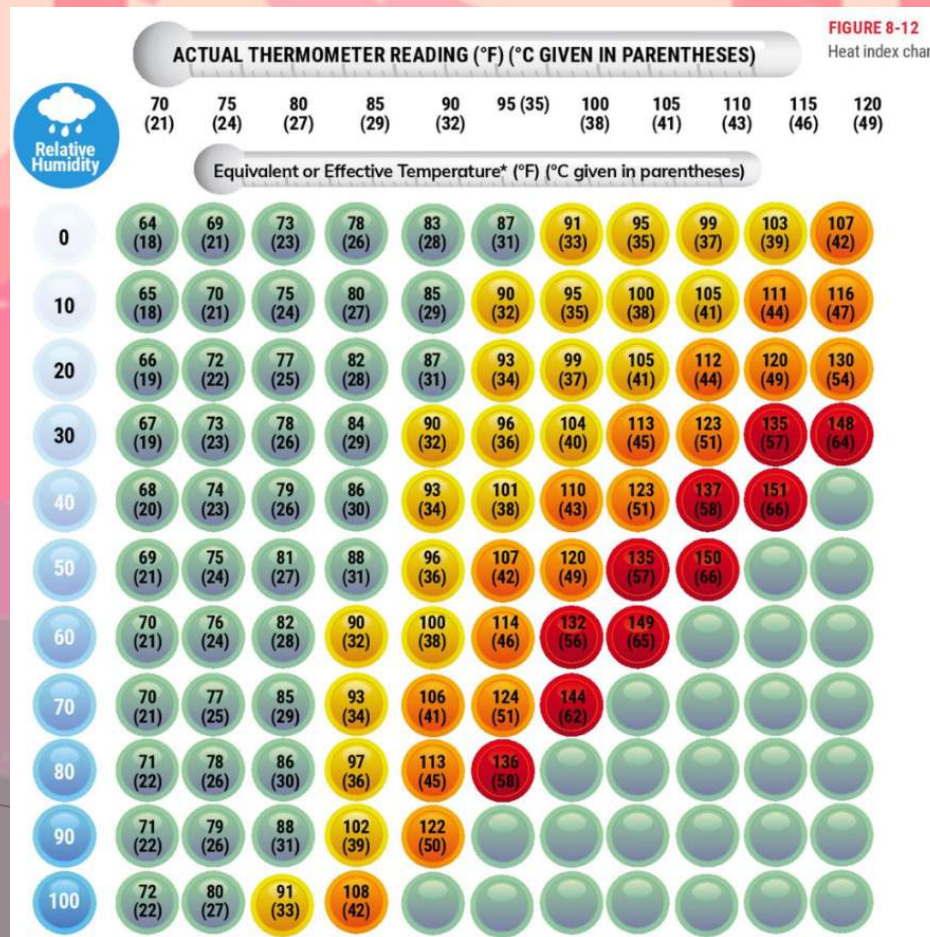


[PG 269-272]

	SIGNS & SYMPTOMS	RECOMMENDATIONS
Heat Edema	Temporary swelling of the extremities	No treatment Support stockings may help
Heat Cramp	Profuse sweating Spasms at arms, legs, or abdominal muscles	Direct pressure to cramp and release Gentle massage Give fluids and electrolytes Passive stretching
Heat Syncope	Decreased SV, Q, BP Syncope	Give fluids Rest with feet raised (Prevention: avoid standing too long in hot environment)
Heat Exhaustion	Weak, rapid pulse, LBP Fatigue, headache, dizziness, nausea, general weakness Paleness Cold, clammy skin, profuse sweating Elevated body core T ($\leq 40^{\circ}$)	Stop exercising Move to a cool, ventilated area Lie down and elevate feet 12-18" Give fluids Monitor T
Heat Stroke	Hot, dry skin, bright red skin color Rapid, strong pulse Labored breathing Change in mental status (e.g. irritability, aggressiveness, or anxiety) Elevated body core T ($\geq 40^{\circ}$)	Stop exercising Remove as much clothing as possible Cool the body immediately in any way possible (wet towel, ice bath, fan, alcohol rubs) Give fluids EMS immediately

EXERCISING IN THE HEAT

[PG 269-272]



Effective Temperature (°F) (°C given in parentheses)	Heat Stress Risk with Physical Activity and/or Prolonged Exposure
90-105 (32-41)	Heat cramps or heat exhaustion possible
106-130 (41-54)	Heat cramps or heat exhaustion likely Heat stroke possible
131-151 (54-66)	Heat stroke highly likely

Figure 8-12: Heat index chart

RESPONSE TO ENVIRONMENTAL FACTORS - HEAT



Begin exercising in
the heat gradually



Wear lightweight, well-
ventilated clothing



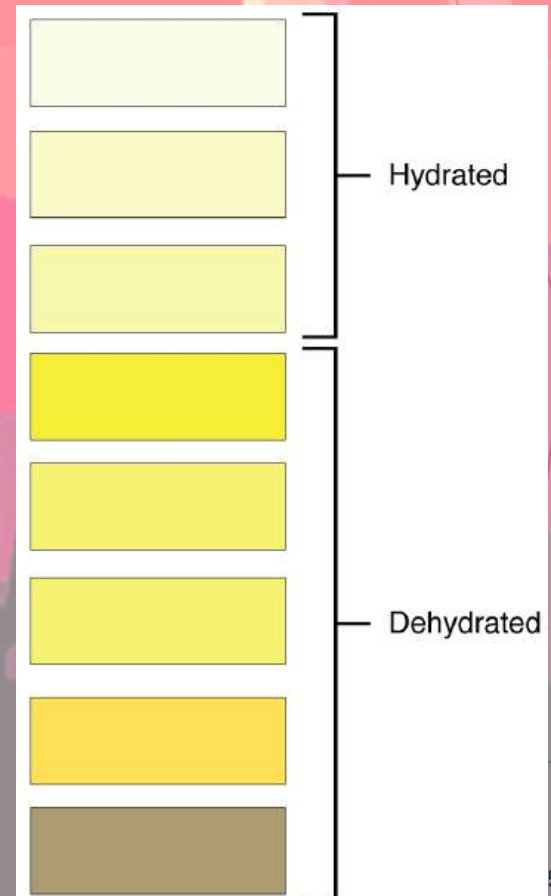
Maintain air
movement



Start the exercise
with hydration



Replace body fluids during
and after the exercise



RESPONSE TO ENVIRONMENTAL FACTORS - HEAT

	Fluid	Comments
Before Exercise	▶ Drink 5–7 mL/kg (0.08–0.11 oz/lb) at least 4 hours before exercise (12–17 ounces for a 154-lb individual)	▶ If urine is not produced or is very dark, drink another 3–5 mL/kg (0.05–0.08 oz/lb) two hours before exercise. ▶ Sodium-containing beverages or salted snacks will help retain fluid.
During Exercise	▶ Monitor individual body-weight changes during exercise to estimate sweat loss. ▶ Composition of fluid should include 20–30 mEq/L of sodium, 2–5 mEq/L of potassium, and 5–10% of carbohydrate.	▶ Prevent a >2% loss in body weight. ▶ Amount and rate of fluid replacement depends on individual sweating rate, environment, and exercise duration.
After Exercise	▶ Consumption of normal meals and beverages will restore euhydration. ▶ If rapid recovery is needed, drink 1.5 L/kg (23 oz/lb) of body weight lost.	▶ Goal is to fully replace fluid and electrolyte deficits. ▶ Consuming sodium will help recovery by stimulating thirst and fluid retention.

Table 8-3: Fluid Replacement Before, During, and After Exercise

RESPONSE TO ENVIRONMENTAL FACTORS - COLD



Hypothermia



Frostbite

RESPONSE TO ENVIRONMENTAL FACTORS - COLD

- Wear several layers of clothing
- Wear breathable clothing
- Allow for adequate ventilation of sweat
- Replace body fluids in the cold, just as in the heat
- Monitor body weight



RESPONSE TO ENVIRONMENTAL FACTORS – HIGH ALTITUDE

Altitude sickness

- Shortness of breath, headache, lightheadedness, nausea

Exercise at altitude

- Not to exercise on the first night
- Prolong warm-up and cool-down
- Start with lower intensity exercise
- Frequent breaks

RESPONSE TO ENVIRONMENTAL FACTORS – AIR POLLUTION

- Urban pollutants: SO₂, CO, NO_x, O₃, etc.
- Suggestions:
 - Exercise early in the morning
 - Avoid high traffic areas
 - Reduce exercise intensity
 - Avoid outdoor training in extreme conditions



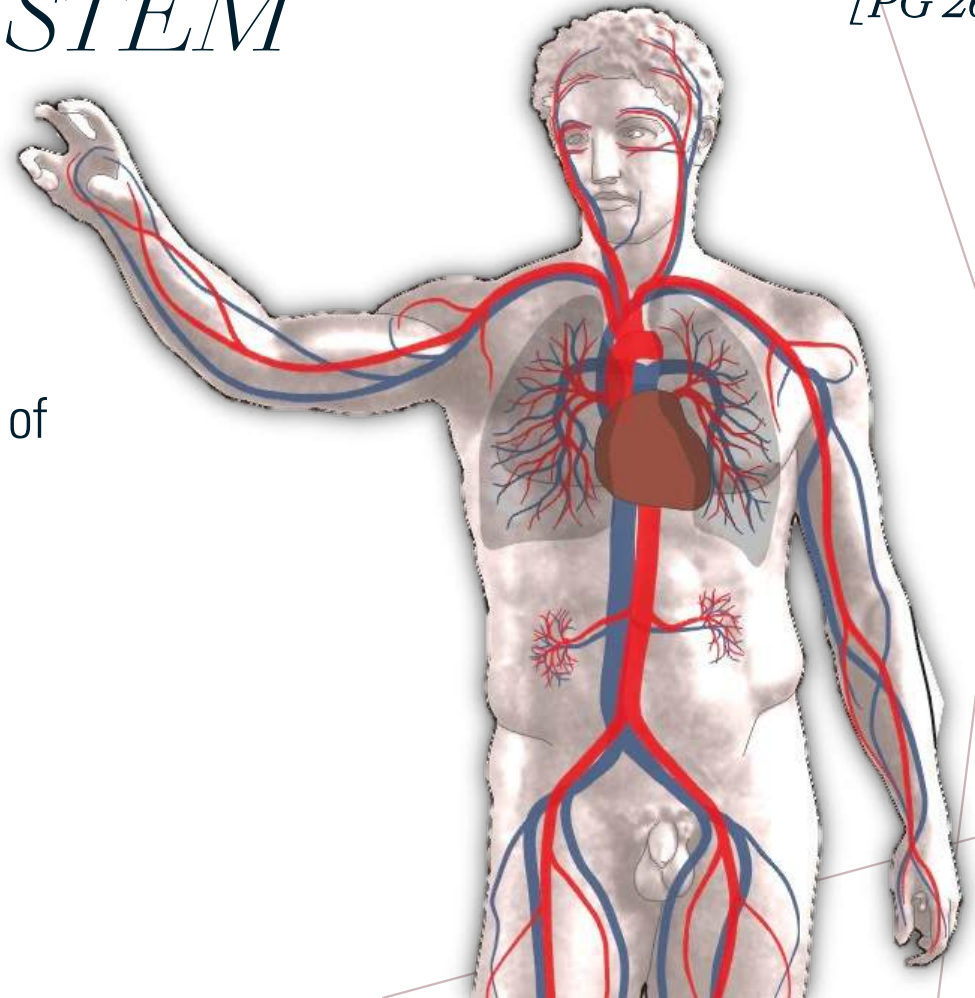
SPORT PRINCIPLE

S PECIFICITY
P ROGRESSION
O VERLOAD
R EVERSIBILITY
D IMINISHING RETURN

S pecific
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I mposed
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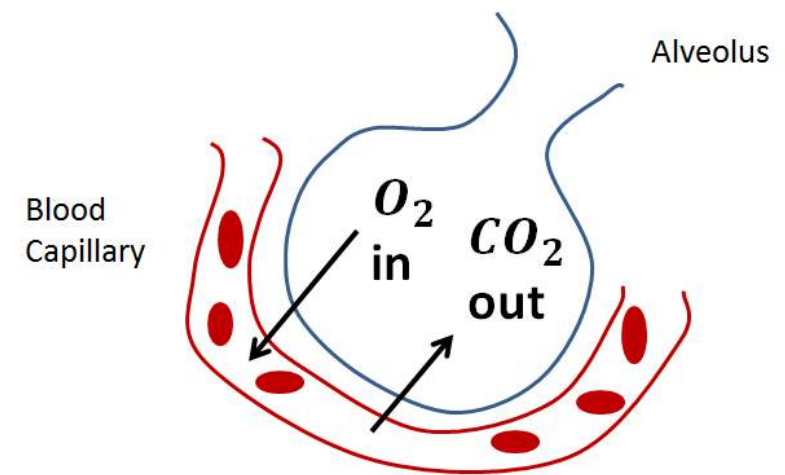
CHRONIC ADAPTATIONS TO CT – CARDIOVASCULAR SYSTEM

- ↗ Maximal cardiac output
- ↗ Stroke volume
- Hypertrophy of left ventricle wall muscles and ↗ of chamber volume
- ↘ HR at rest and during submaximal exercise
- ↗ Capillary density around the recruited muscles



CHRONIC ADAPTATIONS TO CT - RESPIRATORY SYSTEM

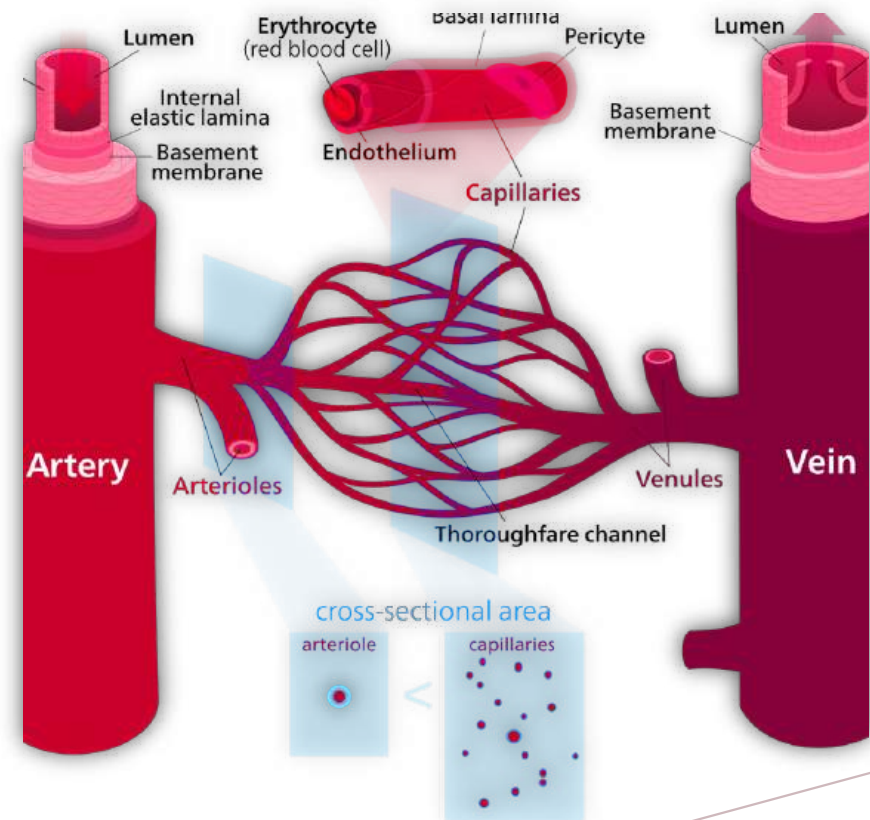
- Submaximal exercise: ↗ tidal volume, ↘ breathing frequency
- Maximal exercise: ↗ tidal volume and breathing frequency
- The respiratory muscles adapt to increase ventilation
 - Diaphragm: key breathing muscles in passive breathing
 - Other muscles of inspiration:
 - External intercostals, sternocleidomastoid, scalene, pectoralis minor, serratus anterior, serratus posterior superior
 - Other muscles of expiration:
 - Rectus abdominis, serratus posterior inferior, internal intercostals



CHRONIC ADAPTATIONS TO CT

– MUSCULAR SYSTEM

- ↗ Hypertrophy of Type I muscle fibers
- ↗ Aerobic capacity of the trained muscles
- ↗ Size and number of mitochondria of the recruited muscles
- ↗ Myoglobin content
- ↗ Number of aerobic enzymes
- ↗ Glycogen and triglyceride stores



CHRONIC ADAPTATIONS TO CT – VO_2MAX , VT

[PG 264-265 & 282]

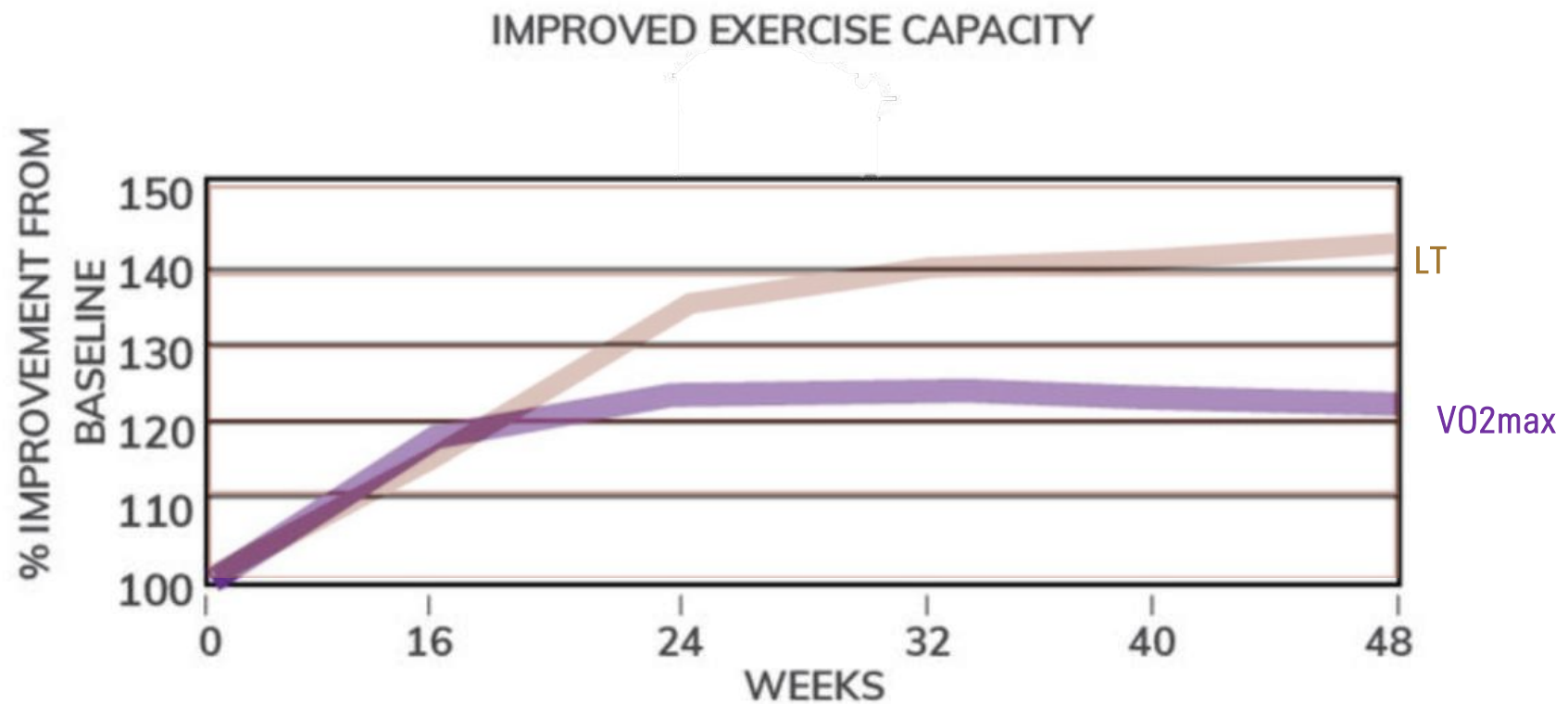


Figure 8.8

FITNESS ASSESSMENTS - BEFORE

[PG 222]

REST	Adequate rest (e.g. 6-8 hrs the night before) No vigorous exercise 24 hrs prior to the test
EAT	Moderate food intake 2-4 hrs prior to the test
DRINK	Adequate hydration
"BREAK"	Free of any injury or related orthopedic problems
MEDS	Abstinence from chemicals that accelerate HR, except for prescribed medications Take note of any medications the client is on
WEAR	Proper attire (loose-fitting, sturdy, tied athletic shoes)
INSTRUCTION	Specific testing procedures and expectations
CONTRAINDICATION	Conditions for terminating a test. Clients are made clear that they may terminate a test for any reason at any time.

FITNESS ASSESSMENTS - TEST TERMINATION

Indications for Terminating Exercise Testing

- Onset of angina or angina-like symptoms
- Drop in systolic BP of >10 mmHg from baseline BP despite an increase in workload
- Excessive rise in BP: systolic pressure >250 mmHg or diastolic pressure >115 mmHg
- Shortness of breath, wheezing, leg cramps, or claudication
- Signs of poor perfusion (e.g. ataxia, dizziness, pallor, cyanosis, cold or clammy skin, or nausea)
- Failure of HR to rise with increased exercise intensity
- Noticeable change in heart rhythm
- Client's request to stop
- Physical or verbal manifestations of severe fatigue
- Failure of testing equipment

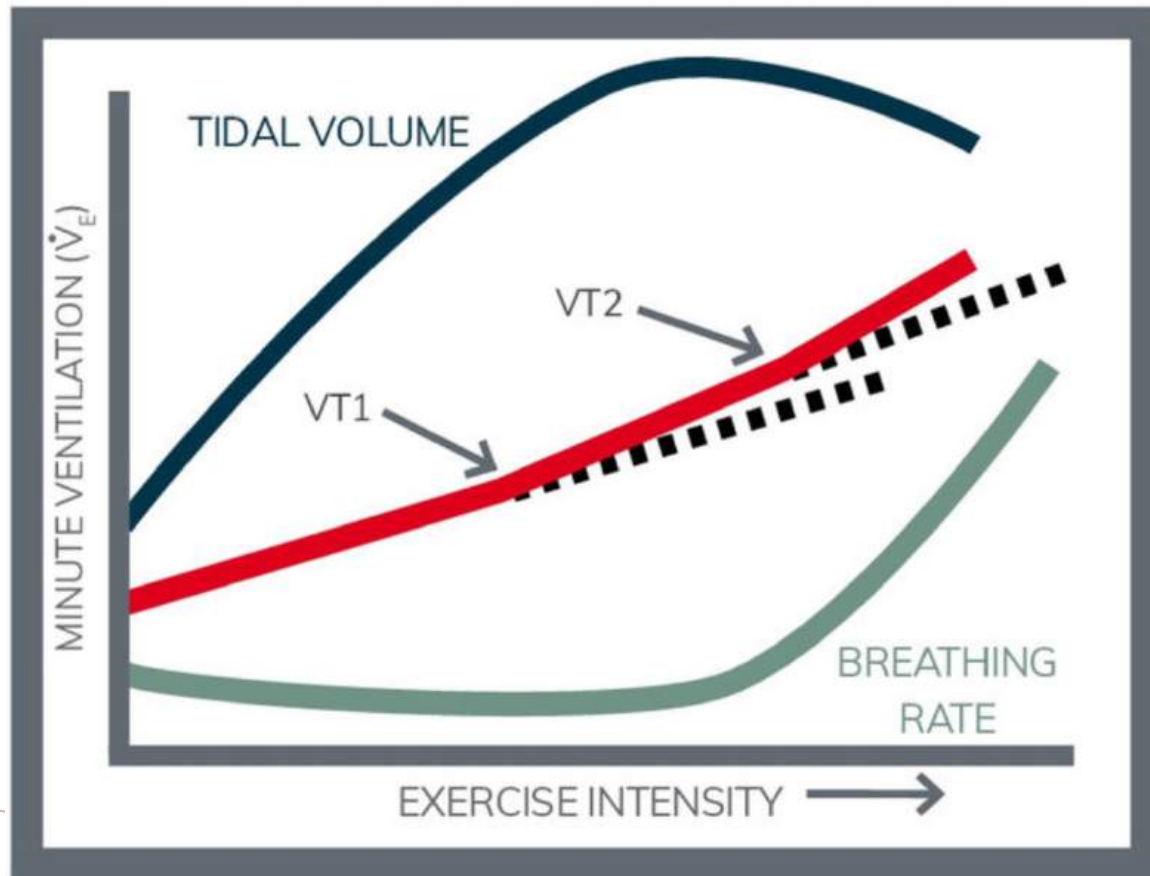


RATINGS OF PERCEIVED EXERTION (RPE)

6	At rest	0	At rest
7	Very, very light	1	Very easy
8		2	Somewhat easy
9	Very light	3	Moderate
10		4	Somewhat hard
11	Fairly light	5	Hard
12		6	
13	Somewhat hard	7	Very hard
14		8	
15	Hard	9	Very, very hard
16		10	Maximum
17	Very hard		
18			
19	Very, very hard		
20	Maximum		

RPE

FITNESS ASSESSMENTS - VT



Note: VT1 = First ventilatory threshold; VT2 = Second ventilatory threshold

Figure 8-19: Ventilatory effects during aerobic exercise

FIELD ASSESSMENTS FOR VTS

[PG 298-299]

Contraindications:

- Individuals with certain breathing problems (e.g. asthma, COPD)
- Individuals prone to panic/anxiety attacks
- Those recovering from a recent respiratory infection
- Individuals who are not fit enough to perform or benefit from the assessment

FIELD ASSESSMENT FOR VT1 – TALK TEST

1. Choose the relevant exercise modality
2. Perform a light warm-up for 3 to 5 min at RPE 2 – 3
3. Adjust the workload to moderate, or RPE 3 – 4
4. Slowly increase the intensity at small increments
5. Measure HR at the last 20 to 30 s of each stage
6. Identify VT1 by asking the client to recite a familiar passage
7. The test concludes when talking is not entirely comfortable
8. Cool down for 3 to 5 min

FIELD ASSESSMENT FOR VT2 THRESHOLD TEST

[PG 301-302]

1. Light warm-up at RPE 2 – 3 for 3 – 5 min, HR < 120 bpm
2. Begin the assessment by ↗ the intensity to the predetermined level
3. Allow the individual to make changes to intensity as needed during the first few minutes
4. Maintain the pace until 20 min is reached
5. Record HR at each min interval during the last 5 min
6. Use the average HR of the last 5 min
7. **[VT2 HR] = [Test HR] x 0.95**

ROCKPORT (1 MILE, 1.6 KM) WALKING TEST

- Developed to estimate $\text{VO}_{2\text{max}}$ for men and women ages 18-69 years.
- Requires only walking at a fast pace, useful for testing older or sedentary clients.
- Instruct the client to walk 1 mile (1.6 km) as briskly as possible.
- Immediately after the test, measure the HR for 15 s and x4
- Estimate the $\text{VO}_{2\text{max}}$ (ml/kg/min) using the equation:

$$\text{VO}_{2\text{max}} \text{ (ml/kg/min)} = 132.853 - (0.0769 \times \text{BW [kg]}) - (0.3877 \times \text{age [yr]}) + (6.315 \times \text{gender}^*) - (3.2649 \times \text{time [nearest hundredth of a min]}) - (0.1565 \times \text{HR [bpm]}).$$

*substitute 1 for males, 0 for females.

1.5 MILE (2.4 KM) RUN

- Measure the time it takes for client to run 1.5 mile
- Instruct clients to cover the 1.5 mile (2.4 km) distance in the fastest possible time. Walking is allowed, but the objective is to complete the distance in as a short amount of time as possible
- Record the time to finish

Use the equation to estimate the client's $VO_{2\max}$ (ml/kg/min): $VO_{2\max} \text{ (ml/kg/min)} = 88.02 - (0.1656 \times \text{BW [kg]}) - (2.76 \times \text{time [nearest hundred of a min]}) + (3.716 \times \text{gender}^*)$.

*substitute 1 for males and 0 for females.

*FIELD ASSESSMENTS OF
AEROBIC ENDURANCE*

ASSESSMENT



EXTRA READING MATERIALS

[1] R. L. Hughson, "Oxygen uptake kinetics: historical perspective and future directions," *Appl. Physiol. Nutr. Metab.*, vol. 34, no. 5, pp. 840–850, Oct. 2009, doi: 10.1139/H09-088.

[2] For aerobic endurance test protocols, go to: [Your ACE account]→"Personal Trainer Resources"